MIL-C-38214B(USAF)
AMENDMENT 3
31 MARCH 1983
Superseding
AMENDMENT 2
19 Aug 1980

MILITARY SPECIFICATION

COMPASS, MAGNETIC, MOUNTED

This amendment forms a part of Military Specification MIL-C-38214B(USAF), dated 7 October 1968 and is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

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2.1: Under "SPECIFICATIONS, Federal", add:

"PPP-B-601 - Boxes, Wood, Cleated Plywood"

*Add the following boxed paragraph to the bottom of page:

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PAGE 2

*Under "STANDARDS, Military", delete "MS33586", change "MIL-STD-100" to read "DOD-STD-100" and add the following:

"MIL-STD-785 - Reliability Program for Systems and Equipment Development and Production
MIL-STD-889 - Dissimilar Metals
MIL-STD-965 - Parts Control Program"

3.1: Delete paragraph and substitute the following:

"3.1 First article. When specified, the contractor shall furnish sample units for first article inspection and approval (see 4.2 and 6.2)."

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3.3.1.1: Delete "MS33586" and substitute "MIL-STD-889".

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*3.4.5.3: Add the following at the end of the existing paragraph:

"The compensator frame shall be made of die cast zinc, cadmium plated. The gears and shafts shall be of one-piece brass construction, cadmium plated. Compensator magnets shall be inserted through the appropriate

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shafts. The cadmium plating of both zinc and brass will provide electrolytic corrosion protection for the dissimilar metals in contact. The compensator shall be able to pass the salt spray test of section 4 of this specification when tested separately (i.e., not mounted in the compass)."

*3.4.6.1, third sentence: Delete third sentence and substitute:

"The prime standard shall be equipped with the applicable lighting, shall meet all specification requirements and shall be used as the reference for quality conformance acceptance of the lighting on production compasses."

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*3.4.10: Delete paragraph and substitute:

"3.4.10 Reliability. The compass shall demonstrate an upper test MTBF (θ_0) of 3000 hours. A failure shall be defined as any degradation below the minimum operational limits specified herein."

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3.7: Delete "MIL-STD-100" and substitute "DOD-STD-100".

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*Add new paragraph as follows:

"3.14 Parts control. A parts control program shall be conducted in accordance with MIL-STD-965."

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4.2,a: Delete "Qualification tests" and substitute "First article inspection".

4.4: Delete "Qualification testing" and substitute "First article testing".

*4.4.2: Delete paragraph and substitute:

"4.4.2 Test report. When the contractor is responsible for testing, the contractor's evidence of testing shall be submitted to the procuring agency for evaluation and shall consist of:

a. Three copies of a test report in accordance with MIL-STD-831 (see Contract Data List).

b. The samples that were tested, in the "as is" condition.

c. Two sets of assembly drawings."

4.4.3: Delete "Qualification tests" and substitute "First article tests".

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*4.5.2.4.1, line 1: Delete "Figure 4" and substitute "MIL-STD-781C, Test Plan IC as modified by the following". Line 9, change "31,400" to read "30,240".

*Add new paragraph as follows:

"4.5.3.2 Failure Reporting, Analysis and Corrective Action System (FRACAS). FRACAS shall be initiated in accordance with MIL-STD-785, Task 104."

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*Figure 4: Delete and substitute new figure 4.

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4.6.29: Add the following before sentence one of existing paragraph:

"Testing shall be in accordance with MIL-STD-781C, Test Plan IC (modified). Modification shall be in accordance with contents of paragraph 4.5.2.4.1 herein."

Lines 2 and 5, existing paragraph, delete "figure 5" and substitute "figure 4". Last line, change "the compass shall" to read "each compass shall also".

*4.6.29.1: Delete "figure 5" in line 2 and substitute "figure 4".

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*Figure 5: Delete figure 5 entirely.

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*4.6.30: In line 2, delete "for a total of" and substitute "until each unit has accumulated at least".

4.7: Delete paragraph and substitute the following:

"4.7 Inspection and test. Tests of methods of preservation and packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this specification."

Delete 5. through 5.3 and substitute the following, also renumber 5.4 to reflect 5.3:

"5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be Level A or C as specified.

5.1.1 Level A.

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5.1.1.1 Cleaning Magnetic compasses shall be cleaned in accordance with Process C-1 of MIL-P-116.

5.1.1.2 Drying. Magnetic compasses shall be dried in accordance with the appropriate process of MIL-P-116.

5.1.1.3 Preservation application. Not applicable.

5.1.1.4 Unit packaging. Unless otherwise specified by the procuring activity, each magnetic compass shall be packaged in unit packs of one each in accordance with Method IA-8 of MIL-P-116. The completed packs shall be overboxed in containers conforming to PPP-B-566, PPP-B-665, PPP-B-676, or PPP-B-636. Cushioning shall be applied between the box and unit container of a type, density, configuration, and thickness that will insure that shock transmission does not exceed the peak G-value specified herein for magnetic compasses when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Same as level A except method III of MIL-P-116 applies in lieu of specified method.

5.2 Packing. Packing shall be level A, B, or C as specified by the procuring activity.

5.2.1 Level A. Compasses packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A and B, Class Overseas, unless otherwise specified by the procuring activity. Insofar as practical, exterior shipping containers shall be of uniform shape size, and of minimum tare and cube, consistent with the protection required."

* 5.2.2 Level B. Compasses packaged as specified in 5.1.1 shall be packed in weather-resistant class exterior shipping containers conforming to PPP-B-636. Other requirements as specified in 5.2.1 are applicable.

5.2.3 Level C. Packing shall be applied which affords adequate protection during shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and reglations and may be the contractor's commercial practice provided the latter meets the requirements of this level."

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*6.2: Add the following subparagraphs:

"g. When first article is required (see 3.1 and 4.2)

h. Any special instructions for qualification of the prime standard(s) for lighting

i. Whether tests of paragraphs 4.5.2.2c, d, e and f may be performed on empty cases. If no statement is made, complete compasses must be used."

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Also add to subparagraphs c and d the following:

"(Sampling plan A is always required)"

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*6.3.2" Delete "minimum acceptable" and substitute "lower test (θ_1)" and delete "contractually specified" and substitute "upper test (θ_0)".

*6.4: Delete paragraph entirely.

*NOTE: The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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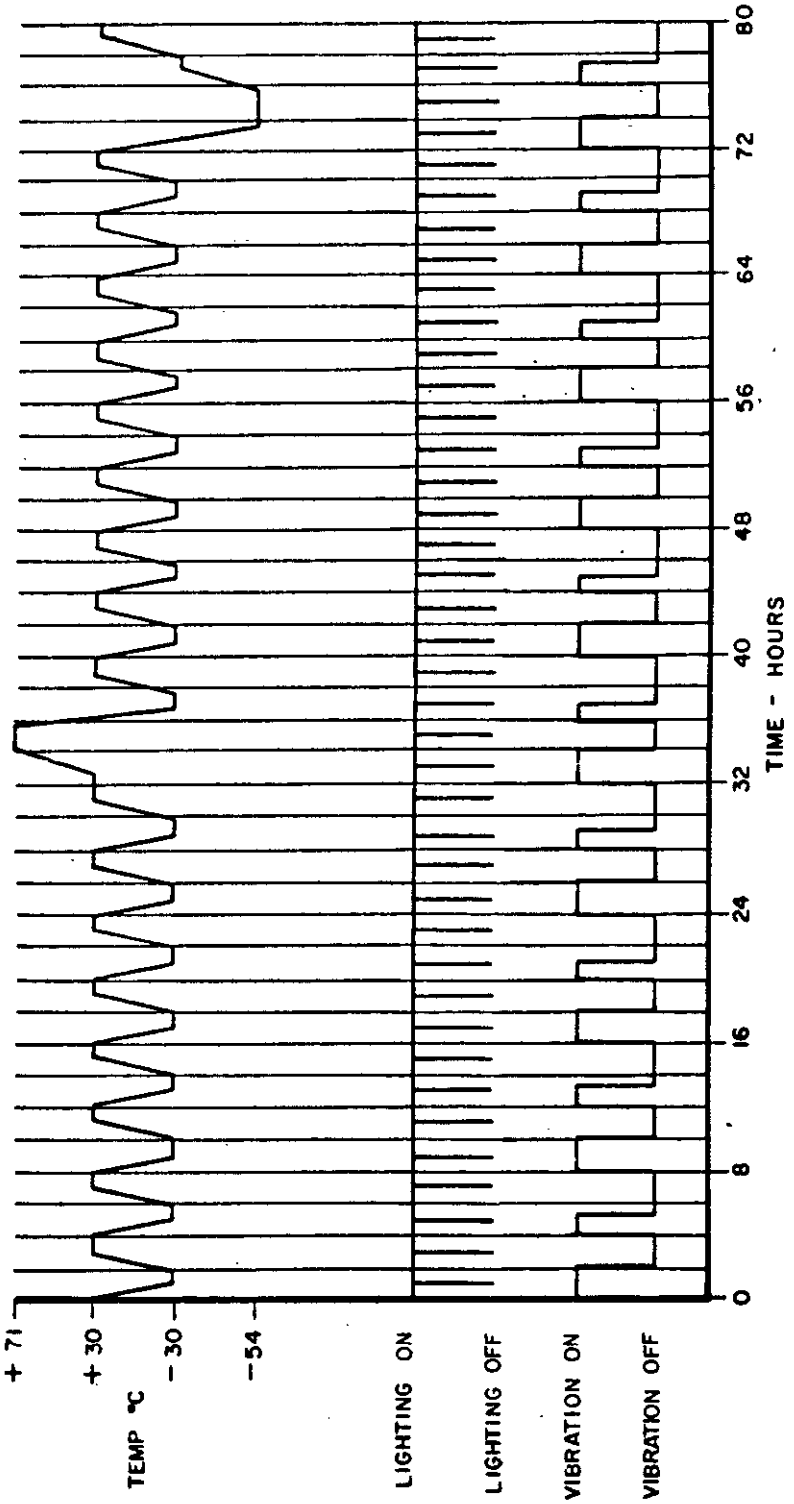


FIGURE 4. Environmental Cycling

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